

SCIENCE · PEOPLE · LANDSCAPES

ADAPT–Nan

Growing knowledge, cultivating change

Advancing Development of Agroecology in Nan Province, Thailand

การพัฒนาระบบนิเวศเกษตรในจังหวัดน่าน ประเทศไทย

Participatory research
Knowledge created together

Experiential learning
Learning through doing

Community engagement
Change grounded in place

Rooted in Nan. Growing together.

Science meets local wisdom

ADAPT–Nan connects researchers, educators, young people and farming communities to explore a transition from chemical-intensive monocultures of maize and rubber toward integrated, sustainable systems, including rice, vegetables, tea, cocoa, coffee and medicinal herbs, that secure food sovereignty. Farmer knowledge is valued alongside scientific approaches, so that knowledge is co-created and put into practice.

Why it matters

- The farming population is ageing: the average Thai farmer is now nearly 60, and few children plan to take over the farm.
- Many farmers remain locked into intensive, chemical-dependent practices — and often into debt.
- Mixed systems integrating trees, crops and livestock are found on about 70% of Southeast Asian farms and underpin rural livelihoods.

Challenges agroecology addresses

Climate — mitigation and adaptation

Water — droughts and floods

Air — biomass burning, haze and PM 2.5

Biodiversity — forests and ecosystem health

Food safety — agrochemicals and safe-food systems

OUR PURPOSE

Support the people shaping agroecology's future

Promote participatory research and transformative pedagogy through co-designed experiential learning.

Re-enchant agriculture for younger generations through agroecological practices grounded in respect for the environment.

How we work



Observe & listen

Direct observation, interviews and focus groups bring different perspectives into the conversation.



Explore together

Role-play games and participatory workshops help people explore complex systems and possible futures.



Learn through practice

Field learning connects scientific ideas with community experience and intergenerational knowledge.

Understanding the land, Supporting the people

Three connected research themes and nine topics explore pathways toward more resilient farming systems and communities.



1 Agrarian change

Agricultural transformations, environmental pressures, and the links between forests and farming.

1.1 Agricultural transformations and environmental issues — drivers of change: droughts and floods, ageing farmers, debt, technologies

1.2 Toxic landscapes and community livelihoods — getting out of agrochemicals through OneHealth, role-play games and scenarios.

1.3 Biodiversity dynamics between forest and agriculture — balancing landscape mosaics: agroforestry, community forests, etc.



2 Social engagement

Organic farming networks, Nan's native red cows, and young people's engagement with biodiversity.

2.1 Organic networks — sustaining smallholder incomes: the future of organic farming, certification and marketing, links with ecotourism.

2.2 Native Red Cows — an action-research to characterise and preserve the Nan red cow landrace through a farmer network.

2.3 La-on Nan — gaming simulations and community forest monitoring that pass forest knowledge on to young people.



3 Agroecology emergence

Learning from local initiatives, civil society and experiential education to support an agroecology movement.

3.1 Learning centers — farms, temples, royal projects and community sites as living laboratories and islands of agroecological coherence.

3.2 From community engagement to agroecology movement — Nan's civil-society and cross-sector collaboration as drivers of change.

3.3 Experiential learning for advancing agroecology — leadership, adaptive spaces and co-creation to tackle persistent "wicked problems".

Agroecology Experiential Learning

The ADAPT BSc Curriculum in Nan Province, Thailand

Curriculum Architecture

45 Hours | 3 Credits

Phase 1 - Pre-Field Preparation

16 Hours, Online

Week 1:
Foundations of Agroecology
(Principles to transitions)

Week 2:
Sustainable Practices
(Soil, water, and IPM)

Week 3: Systems Thinking
(Companion modeling, "Wicked Problems")

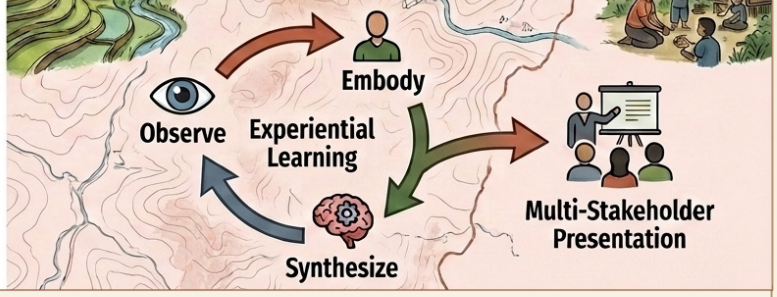
Week 4:
Community Solutions
(Business models & food systems)

Phase 2 – Intensive Field Block

29 Hours, Nan Province

7 Days of Immersive Learning

Based in Nan with daily excursions.
Structured entirely around the **Experiential Learning (Observe → Embody → Synthesize).**
Culminates in a multi-stakeholder panel presentation.









A higher education curriculum focused on agroecology and sustainable agriculture and food systems through an immersive study program in Nan Province.

The course integrates online preparatory seminars with a seven-day intensive field block where students engage directly with local farmers and diverse agricultural landscapes.

The course aims to transform students into skilled facilitators and environmental stewards.

By utilizing participatory tools and systems thinking the program teaches students to navigate complex socio-ecological challenges and to codesign sustainable business models and agroecological solutions with local communities.

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Chula Chulalongkorn University

CSE

IRD Deploying research Sharing science Transforming the future
French National Research Institute for Sustainable Development

Kasetsart University

Nan Community College

Rajamangala University of Technology Lanna Nan

ADAPT-NAN AGROECOLOGY CURRICULUM

Field Block Guide

Seven days of immersive learning in Nan Province, Thailand

9 – 15 November 2026

ADAPT-Nan
Advancing Development of Agroecology in Nan Province in Thailand

Intensive Field Block

Based in Nan town with daily excursions

The field block brings the online preparation to life. Each day you will work alongside farmers, monks, entrepreneurs and community organisers who have built agroecological alternatives in Nan's watersheds, and learn to trace the links between climate change, food security and rural livelihoods. The week ends with your own proposals, presented to a panel of local stakeholders.

7	5	3	29
days in Nan	host sites	groups per site	field hours

A day in the field

Observe
Hosts share their story and the site
≈ 40 min

Embody
Hands-on practice in small thematic groups
1–2 h

Synthesize
Debrief and exchange across groups
≈ 40 min

Every group studies one theme in depth, then shares it with the others.

Our base in Nan

ศูนย์พันธกิจเพื่อสังคม
CENTER FOR SOCIAL ENGAGEMENT
Chulalongkorn University

Before you go

- **Complete** the 4 online modules on Google Classroom (16 h).
- **Weather:** mild days, cool evenings; bring a jacket and a hat.
- **Field wear:** comfortable clothes and closed shoes.
- **Water:** refill tanks on site, so bring your own bottle.
- **Mosquitoes:** nets are provided; bring repellent as well.
- **Farmer School (9–10 Nov):** separate dorms for women and men.
- **Temple visit (11 Nov):** shoulders and knees covered.

Where we go



- **9** 9 Nov **Chumchon Ton Nam Nan (CTN)** · Tha Wang Pha
- **10** 10 Nov **Organic rice farm & Royal rice mill** · Tha Wang Pha
- **11** 11 Nov **Pong Kham Community Learning Center** · Santi Suk
- **12** 12 Nov **OrigiNan Space** · Mueang Nan
- 13 Nov **Design thinking workshop (base)** · Mueang Nan
- **14** 14 Nov **City visit · Nan old town** · Mueang Nan

Site locations are indicative. District boundaries: OpenGISData-Thailand.

Day by Day

Arrival & orientation briefing: Sun 8 Nov, CU-Phasing Center · Field days 9–14 Nov · Departure 15 Nov

MON 9



Chumchon Thon Nam Nan (CTN)

Farmer School

Nan Agri-Nature Centre, Tha Wang Pha · Hosts: Kul Panyawong (founder) & Khun Pui (coordinator)

Pa Kul introduces why ChTN was founded and how self-reliance responds to climate change and food insecurity. Three groups then explore one theme each: **food production & composting**; **water management (building a check dam) and the 5-level forest**; **food processing & marketing**. Afternoon debrief and cross-group exchange.

TUE 10



Organic rice, from seed to mill

Farmer School

Community enterprise for organic rice & Royal rice mill, Tha Wang Pha · Hosts: Pho Chin & Pho Chot

Morning in the paddies near harvest time: **seed management, organic rice and rice–fish systems, single-seedling transplanting, variety selection**. Lunch with village households, tasting their own organic rice. Afternoon (14:00–16:00) at the royal rice mill: milling, packing, waste management, and coping with floods and an ageing workforce.

WED 11



Pong Kham Community Learning Center

CU-Phasing Center

Wat Pong Kham, Santi Suk · Host: Phrakhru Sujinnanthakit (Phrakhru Somkid)

A temple-led PGS organic network that shifted local farming from a capital base to a dharma base, with a monk doing the marketing. Groups: **integrated farming on member farms**; **forest & herbal processing**; **supply chain: sorting, packing, transport**. Afternoon dialogue with the network of development monks.

THU 12



OrigiNan Space, a hub of social innovation

CU-Phasing Center

Mueang Nan · Producer cluster linking upstream, midstream and downstream

Morning groups analyse how entrepreneurs connect farm resources to market demand: **Baan Piang Tawan & Baan Tua Lisong** (agri-products); **Luk Chan PGS** (farm systems); **Cheewawithi Nam Kian** (herbal, natural living). Afternoon at OrigiNan Space on cluster management and social innovation.

FRI 13



Design thinking workshop

CU-Phasing Center

CU-Phasing Center · Mentor: Khun Khem (Kai Pa Camp, Hug Muang Nan Foundation)

Groups turn the week's field insights into **co-designed agroecological solutions and business models**. Khun Khem, whose bamboo farm, craft space and community market offer local alternatives to monocropping, mentors the teams as they test their ideas.

SAT 14



Presenting back · Nan old town

CU-Phasing Center

CU-Phasing Center (morning) · Nan old town (afternoon)

Morning: **multi-stakeholder panel presentation**, summary and closing session with hosts and partners. Afternoon: visit Nan old town Featuring UNESCO Nan creative city.

SUN 15

Departure for Bangkok

Safe travels, and thank you to all our host communities.

ADAPT-Nan — Advancing Development of Agroecology in Nan Province in Thailand

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Nan Headwaters Community Learning Center

Chumchon Ton Nam Nan (ChTN) · Nan Agri-Nature Center · Tha Wang Pha, Nan

Soil, water, forest and people — restoring a headwater landscape through natural agriculture

Background

Nan Province is more than 80% mountainous and only about 10% of its land is arable. Most of that farmland has been given over to monocultures of cash crops such as rubber and maize, setting off a vicious circle: deforestation brings soil erosion and landslides, agrochemicals wash into the waterways, and the cycle begins again. The stakes reach far beyond Nan. Its river supplies about 47% of the flow of the Chao Phraya, the “River of Kings” that irrigates central Thailand’s rice bowl, and as upstream forests disappeared, floods and droughts followed downstream.

The Nan Headwaters Community learning center, also known as the Nan Agri-Nature Center, was founded in Tha Wang Pha District in 2013 in response. Its guiding idea is that Nan’s problems are national, not merely local, and that even a small demonstration site can spread practices outward. Drawing on King Rama IX’s philosophy of soil, water, forest and people and on the Sufficiency Economy Philosophy, it shows how degraded farmland can become a diverse, water-retaining landscape that provides food security, income from surplus produce and richer biodiversity.





Origins of the Learning Center

Mrs Kul's path back to Nan

The center's founder, Mrs Kul Panyawong, has worked in the environmental field for some twenty years. A former mover and shaker in Bangkok's advertising industry, she spent five years at J. Walter Thompson, then worked in community relations at CHR Community Service. A keen cyclist, she founded a mountain-biking company, ran conservation tourism, and produced travel books, magazines and television programs.

She then spent six years at the Thai Forest Conservation Foundation in Chiang Mai, a well-resourced organization funded by the electricity company EGCO. Over those years she felt increasingly drawn back to her home province, where forest cover was shrinking fast. Returning seemed difficult, though: other organizations were already active in Nan, and she had no local network of her own.

Learning from Ajarn Yak

The turning point came when she met Ajarn Yak (Dr. Wiwat Salyakamthorn), chairman of the Agri-Nature Foundation. He argued that Nan's problems were national in scale, and that a small demonstration point spreading ideas outward could help address them. From him she learned how to put the Khok Nong Na model into practice. She then replicated the model on her own land, refining her skills before passing them on through training at the center she founded.

Overcoming obstacles

Her first attempt to change farming in Nan was to raise funds to pay villagers to farm without chemicals. It did not work: money, she quickly learned, is not the only factor in play. Her new plan was to show people that the system really works.

> ***"Lasting change comes when people actually see you do something and succeed at it. When they see that the Khok Nong Na model works, that's when things start happening."***
>
> *Kul Panyawong, founder of ChTN*

For many farmers, watching their livelihoods improve alongside biodiversity was a revelation. "They see that the land can be a source of food, where they can plant rice, grow vegetables and raise animals," Kul says. "Everything will become more fertile, the soil, water, forests. They will have more food and make more money. Ultimately, their lives will be more stable."

The model reaches the prisons

The same model has since been taught at its smallest scale in prisons, under King Rama X's Khok Nong Na Nam Jai Haeng Khwam Wang ("Hopeful Heartland") project. During COVID-19, a 14-day program was run in 137 prisons nationwide, including in Nan, for inmates facing unemployment on release, most of them from farming families. It began at Klong Prem Prison on a plot of just 12 × 11 meters and covered the Sufficiency Economy Philosophy, natural-agriculture principles, land design and soil improvement.

KHOK NONG NA LAND ALLOCATION

30%	30%	30%	10%
water	forest	rice fields	residence

The Site: Soil, Water and Forest

A landscape designed for learning

From the outset Khun Kul designed ChTN as a training center with multiple learning stations, working with a dean of the Faculty of Architecture at King Mongkut's Institute of Technology Ladkrabang. The layout follows King Rama IX's New Theory and the Khok Nong Na model: khok means "mound", nong "marsh" and na "rice field".

THREE FORESTS, FOUR BENEFITS

The integrated-farming system popularized by King Rama IX combines three kinds of "forest": fast-growing trees such as neem and bamboo; vegetables, fruits and herbs; and commercial timber. Together they bring four benefits: wood to build homes and furniture; food and medicine; raw materials for household products such as firewood, kitchen utensils and soap; and shade to protect fields, animals and buildings.

Healing contaminated land

The land had previously been rubber and longan orchards, heavily treated with chemicals, and the 14-rai entrance area was a pesticide-intensive maize field. When Khun Kul arrived, soil tests found arsenic and significant chemical residues across the whole 50-rai property.

She remediated the soil with locally abundant organic materials (leaf litter, rice straw, bamboo and rubber-tree leaves, dry compost and corn husks) and developed seven herbal fermentation formulas for soil improvement, pest repulsion and root nourishment. Soil is prepared in five steps: dig, break up, mulch, feed with dry and liquid fertilizer, then leave to ferment for a week before planting. The process also lends itself to science lessons on fungal communities.

Keeping every drop of rain

"In Nan, some 87.2 percent of the mountainous terrain consists of upstream forests that are crucial to the flow of the Chao Phraya," Kul explains.

The Khok Nong Na model manages water storage using natural-farming principles, a win-win in which both forests and farmers benefit. Three principles guide the site: retain water in the soil, capture 100% of rainfall, and build biodiversity, the central pillar of ChTN's work.

The paddies were dug into laterite mixed with gravel using traditional techniques: bunds raised with the excavated soil, irrigation channels along their base, vegetables on the bund tops, and fish raised in the fields with sawian, bundles of dry leaves and manure submerged to grow plankton as fish food. Results were visibly better than on neighboring plots, and students from Rajamangala University of Technology Lanna (RMUTL) Nan came to study the rice-growing process here.

The five-level forest

Starting from bare land, Mrs Kul planted trees across five vertical strata:

1. **Tall trees —** teak, makha, phayung
2. **Mid-level fruit trees —** longan, lychee and more than 30 other species
3. **Understory —** cocoa, coffee
4. **Low and creeping plants —** medicinal herbs, pumpkin, pepper
5. **Roots and tubers —** yams and other underground crops

Elsewhere on the site, an orchard due to be converted from longan to rubber was instead left to regenerate. Within a year of stopping herbicide use, 22 tree species had returned on their own from surviving roots, rhizomes and seeds, illustrating King Rama IX's idea of "planting forest without planting". Bird diversity has risen from about 20 species before Mrs Kul arrived to more than 100 today, in keeping with the center's principle: "nourish the soil so trees grow; let trees nourish people."





Activities and Training Content

Three training modules

Mrs Kul uses different tools for different groups in a three-pronged education plan:

1. **Field observation —** for visitors who want to see what happens on the ground; the aim is a change of mindset by learning through seeing and doing.
2. **Training course —** five days and four nights of living, eating and learning according to the Khok Nong Na model, networking with like-minded people.
3. **Volunteering —** course graduates stay at least two weeks to learn to create a Khok Nong Na suited to their own means, and to spread the approach to others.

Courses cover land design and site planning, compost making, household products such as multipurpose cleaning liquid, and traditional health practices.

Circular living and zero waste

Whether people come as students, volunteers or observers, Mrs Kul wants them to grasp the value of circular living. All the food served at the center is grown on site, and leftovers are composted into organic fertilizer. Catering is planned to create no waste: menus are simple dishes made from local ingredients such as fish from the pond and organic vegetables from the garden, packaging is made from natural materials such as banana leaves. Kitchens cook with bio-charcoal, lights run on solar cells and water is circulated by solar-powered pumps, costly to install but worth it for the environment in the long run.

Self-reliance, step by step

The center follows King Rama IX's nine-step ladder: first become self-sufficient in daily necessities (food, medicine and household products), then move on to surplus, sharing and finally production for sale. Khun Kul encourages visitors to aim for at least 25% self-reliance in everyday items such as soap, shampoo and dishwashing liquid. ChTN's own products include organic kaffir-lime-leaf shampoo, turmeric and centella soaps, soy sauce, a "green oil" herbal balm and activated charcoal from branches for cosmetics; during COVID-19 the center produced herbal medicine tablets.

A refuge in times of disaster

ChTN is designed to shelter 300 people for a month in the event of a natural disaster and to serve as a regional relief-distribution hub, with space for tents, food stocks and herbal medicines.

TIMELINE SUMMARY

2013
B.E. 2556
ChTN founded on a 50-rai plot; soil restoration begins.

2016
B.E. 2559
Training programs start generating income.

2018
B.E. 2561
Mrs Kul presents ChTN's water work at the UN on SDG 6.

2020–21
B.E. 2563–64
COVID-19: closed to visitors; 70,000 herbal tablets produced.

2022
B.E. 2565
Featured in Thailand's BCG book for the APEC summit.

2024
B.E. 2567
Bike For Earth hosted in June; Mrs Pui settles at ChTN

2025
B.E. 2568
Site renovated; Araya Camp and Forest Restoration trainings.

2028
B.E. 2571
Goal: ChTN 100% self-financing from its own organic products.





NAN PROVINCE THAILAND



Rice Farmer Field School

Silaphet Sub-district · Pua District · Nan Province

Learning rice, health and Tai Lue life — in the paddies, alongside local farmers

Background

The traditional farming system in Silaphet sub-district involved the cultivation of paddy rice followed by a variety of vegetable crops after the rice harvest, grown for household consumption. Between 1987 and 1992, farmers in the area moved from subsistence into commercial crop cultivation, particularly chili, a shift driven by middlemen who promoted chili cultivation and offered good cash income. On the other hand, the shift from rice transplanting to direct seeding as a labor-saving practice generalized in the area, as in other provinces of the country engaged in a general process of agricultural intensification.

The new cultivation practices led farmers to invest in more chemicals, especially herbicide and insecticides in former "organic by default" rice fields. Problems of decline in soil fertility, yields and crop failures were brought to the attention of the sub-district (tambon) agricultural office (TAO), and experts suggested substituting chili with other crops: peanut, soybean, cucumber, Chinese cabbage, pea and broccoli, to sustain yields and prevent disease. However, the lack of collective crop-selection decisions caused marketing problems, with some crops in over-supply and others in short supply.

Vegetable farmers therefore registered with the TAO to obtain advice on crop types and farm planning. Vegetable farmers in Silaphet established a 'safe vegetables producing group', supported by the sub-district hospital (which carried out random blood tests for chemical residues) and the local agricultural office (which advised on safe and appropriate chemical use). This collective brought the issues of safe food production and consumption to the attention of both farmers and consumers who wanted to reduce chemical use.



Origins of the Farmer Field School

Rice curriculum in Ban Thoung Khong

In 2011–12 (B.E. 2554–55), following a Ministry of Education directive for schools nationwide to develop curricula adapted to local situations and linked to community development, Mrs. Jongrak, then teaching natural sciences (biology) at Sattha Silaphet School, began building a curriculum around rice cultivation. She was in contact with the Joko community center, which worked with rice farmers on seed production, and together they developed activities in Ban Thoungkhong, Tambon Yom, Amphoe Thawangpha, leading to a farmer field school gathering 15 farming households. The approach echoed the King's philosophy of sufficiency economy, and Mrs. Jongrak joined the group to provide training support for her high-school students.

Together, farmers and students worked on seed saving, selection and demonstration of rice varieties (both sticky and non-glutinous), and on the study and conservation of local rice varieties with the support of the Hug Muang Nan foundation. Students often joined activities outside official class hours. The natural-science curriculum, built around rice as a study topic, was developed jointly with farmers, fully adapted to the local context, and evolved as farming practices in the area changed. It was taught directly in farmers' fields, such as Mrs. Phin's farm, with support from the Tambon authorities and the local health station in Silaphet, where Mrs. Suttana Pathamawat and her husband worked.

Mrs. Suttana's public health work

Mrs. Suttana and her husband, Mr. Sathian, were public health workers and academics at the sub-district health station, investigating rising rates of diabetes, hypertension, cancer and other non-communicable diseases. As part of this work and her MSc study, Mrs. Suttana tested villagers' blood cholinesterase levels, an enzyme test used to assess poisoning from the organophosphate and carbamate pesticides commonly used by farmers and others regularly exposed to agricultural chemicals.

Testing was carried out in every village across the sub-district, and it revealed very high exposure rates — the more they tested, the more cases they found. Wanting to understand the underlying cause, Mrs. Suttana launched a campaign against chemical use. At the time, she owned rice land with her siblings but did not farm it herself; others farmed it, using agrochemicals. Villagers pointed out the contradiction: "She campaigns against chemical use, yet her own land still uses chemicals"; so Mrs. Suttana decided to farm the land herself, without chemicals.

> ***"She campaigns against chemical use, yet her own land still uses chemicals."***
>
> What villagers said was the turning point that led Mrs. Suttana to farm her land without chemicals.

Founding of the Rice Farmer Field School

Once Mrs. Suttana began farming the land herself, Mrs. Jongrak asked to use it as a learning space for school children to learn about farming, and the two women worked together to design a curriculum. Mrs. Jongrak's original native-rice curriculum, developed in Thung Khong village (Yom sub-district), fell outside Silaphet's jurisdiction, so the Sila Phet sub-district administrative organization (อบต., OBT) could not support the children studying there. In 2012, the class moved onto Mrs. Suttana's land in the neighboring village of Nakham in Silaphet sub-district, within the same jurisdiction as Mrs. Jongrak's Silaphet high school, and serving as a model for villagers alongside her own farming experiments. With OBT support, an organic market was created and the curriculum evolved to cover the full cycle of rice production, from nursery to harvest, processing and marketing, including training courses on weed and water management.





The curriculum grew from one classroom to two, then three, and the Farmer School was born, initially using roughly 5 rai of paddy land inherited by Mrs. Suttana together with her siblings, Mrs. Kanika and Mr. Sanith, who agreed to make part of the land available for student learning. Mrs. Kanika taught students the rice-farming process, and part of the harvest was used in a curriculum where students learned to process rice into sweets and to sell products through online and offline marketing. Guest accommodation followed in 2012–13 (B.E. 2555–56), built to generate income for the learning center; the "farm stay" accommodation was registered as a commercial business, while the Farmer Field School itself was registered as a learning center.

Training activities then expanded to safe vegetables grown after rice during the winter season: Mo4-grade students studied vegetable crops while Mo5-grade students worked on rice, including traditional variety conservation and labor-saving production techniques. Mrs. Suttana also traveled to Sing Buri province to obtain Riceberry rice seed, a low-sugar content variety, for testing. Students collected water and soil samples to measure chemical contamination and tested agricultural products and blood cholinesterase levels, to better understand the relationship between environmental contamination and human health. Farmers engaged in clean vegetable production to sell to the local hospital, certified by the hospital laboratory, which checked for contamination. Mrs. Suttana used the Farmer School as the subject of her MSc thesis, running training sessions, managing her public health work, and the school simultaneously. The school also began welcoming visiting learners and tourists, both to expose them to the project and to generate supplementary income.

In 2019 (B.E. 2562), Mrs. Suttana and Mr. Sathian resigned from their jobs to run the Farmer School full-time while keeping a health focus, after the rice season they grew vegetables on the same land, tested them, and sold them to Pua Hospital to support villagers' income. Asked how confident she was that homestay income could sustain their livelihood, Mrs. Suttana explained that the school did not require a large budget, so she felt able to manage, and believed tourism income would be enough to cover living costs.

TIMELINE SUMMARY

2011	2012–13	2013–18	2019
B.E. 2554	B.E. 2555–56	B.E. 2556–61	B.E. 2562
Farmer School founded (no proper school building yet, just a small room).	Guest rooms were built; began accepting tourists.	Farmer School in operation, while Mrs. Suttana still worked in public health.	Mrs. Suttana and Mr. Sathian resign from their jobs; income relies mainly on tourists.

Activities and Training Content

Rice cultivation curriculum

Before seedlings are grown, students learn to select seed varieties and test germination rates, then work alongside villagers doing the same tasks, learning directly from local farmers. Students practice meticulous rice-planting techniques, using straw thread to measure precise spacing and tying exact knots, learning the traditional planting process alongside academic principles such as how many tillers a single rice plant produces. This served as a model to show farmers that fewer seedlings can be used, both to save costs and to increase yield.

The curriculum was designed for children who attend regular school daily and often have extra tutoring on weekends, yet, despite being the children of rice-farming families, rarely had the chance to learn farming skills or how to use their own family's fields. A curriculum directly teaching these skills was designed to fill that gap, built to keep teaching costs low by relying on local people as instructors rather than paid outside teachers.





Vegetables are also grown after the rice season: chili, shallots, garlic, lettuce, Chinese cabbage, broccoli, using safe-vegetable cultivation principles. The project has encouraged the wider Silaphet community to grow safe vegetables too, through a community 'Health Charter' that sets quality standards. Produce is consumed within households, sold through middlemen at roadside stalls, and supplied to Pua Hospital, with quality inspected and certified as verified-safe by the Silaphet sub-district administration (OBT), giving consumers confidence in the safety of Silaphet produce.

Homestay development

OBT support for clean food and chemical-free agriculture, alongside other youth-education topics such as drugs and smoking, helped the project in its early years, but the economic model was not sufficient to sustain the free service provided to students. Mrs. Suttana began building houses to create a homestay that would generate resources to support school activities, building house after house since 2013. The homestay now comprises 10 houses of 2 to 6 occupancy each, for a total capacity of 40 people.

Visitor numbers are roughly 30 to 50 people per visit, with about 10 delegations per year, totaling roughly 300 to 500 visitors annually. Uncontrollable external factors affect the business, such as the flood season, when booked guests cancel and request refunds, which the school must honor. Low season for lodging generally runs from March to May, coinciding with the local haze season, when hot weather discourages tourists.

The Farmer School has air conditioning in only one building. However, the school wants to preserve the authentic character of a 'Farmer School' experience close to nature, showcasing Tai Lue culture. Buildings feature many windows and doors for airflow, natural-material ceilings, and gaps in the walls for ventilation. Installing air conditioning would require full renovation, including replacing all ceilings, which the family is reluctant to do given the cost. As a result, the school closes for four months, March–June, generating no income while monthly operating costs continue regardless; during the 8 income-generating months, they must save enough to cover the 4 closed months, managing the whole year's budget around this cycle.



Competition has increased over time — there are now many more accommodations and more diversified tourism options — and repeat visits are rare, with most tourists who come once not returning, further reducing visitor numbers over time.

Business Model and Networking

Funding sources and financial estimates

In the early period the Farmer School had no external budget; all administrative funding came from Mrs. Suttana's personal savings. Later, Mrs. Jongrak sought some funding to help cover snack/meal costs during weekend art classes, though weekend art teaching remained funded largely from Mrs. Suttana's own pocket. As the school expanded from one to three classrooms, costs rose, prompting the construction of guest accommodation to generate income, with guests sometimes donating toward meals and snacks for the children. Income from the accommodation business mostly went back into administration and ongoing construction of new buildings, since only two guest buildings existed initially and construction costs for new units were high. Approximate funding breakdown: personal funds, 60%; tourist income, 40% (split into 20% for running activities/services and 20% for new construction).



Organic Rice Network and Royal Rice Mill

Si Phum Sub-district · Tha Wang Pha District · Nan Province

From flood relief to organic rice — a learning journey

Background

Si Phum sub-district lies on the flat, irrigated floor of the Nan River valley in Tha Wang Pha district is one of the main rice-growing areas of Nan province. Farming families traditionally grew paddy rice for household consumption and sold any surplus, alongside fruit orchards (orange, litchi, longan), upland rice and cash crops such as tobacco. As elsewhere in Thailand, the move to commercial farming brought heavier use of agrochemicals: orchards gave high yields, but at a high cost to health and household budgets. Farmers who sold paddy depended on middlemen and private mills, and faced depressed prices and rising input costs.

At provincial level, Nan's farmers were caught in a vicious cycle of growing debts, overusing agrochemicals and deteriorating health. As farmland shifted to non-food cash crops, the province increasingly brought in food from outside, of unknown quality and unpredictable quantity; a real concern for food safety and food security.

The turning point for Si Phum came in 2006 (B.E. 2549), when flash floods and the overflowing Nan River struck northern Nan. Homes and farmland were badly damaged, and in Tha Wang Pha many farmers could not harvest their crops at all. The response to that disaster laid the foundations of both the community rice mill and, a few years later, the organic rice network.



Origins of the Mill and the Organic Network

A royal rice fund after the 2006 flood

After staff of the Private Secretary's Division of HRH Princess Maha Chakri Sirindhorn delivered royal relief supplies to flooded villages, they reported the scale of crop losses and proposed a **Royal Rice Fund**. The Princess granted money, milled rice and rice seed to start the fund, so that affected households could replant and the community would be better prepared for future disasters.

Mr. Son Khamin, then sub-district headman (kamnan) of Si Phum, played a key role in brokering this support and in persuading villagers to pool their own savings alongside it. Households that received royal seed later returned part of their harvest to the fund, so that other communities could benefit in turn.

From a rice fund to a community mill

Seeing how well the community managed the fund, and responding to villagers' wish for their own mill, the Princess granted a community rice mill. A CPR 1000 community-scale milling unit was donated through the Charoen Pokphand (CP) Group; the site, on high ground safe from floods, was secured with support from the Treasury Department and the Nan Provincial Administrative Organization; and the Si Phum Sub-district Administrative Organization initially paid the mill's staff. The **Royal Rice Mill** (โรงสีข้าวพระราชทาน) was formally opened by the Princess on 27 February 2009 (B.E. 2552), when she symbolically packed the first bags of rice.

The Nan Provincial Cooperative Office acted as mentor: it introduced the cooperative model and trained staff who had no prior background in milling, paddy purchase, accounting or marketing. The mill follows the Sufficiency Economy philosophy and the motto "Nan rice for Nan people". On a royal visit in 2010 (B.E. 2553), Mr. Son, then chairman of the mill, asked for help with irrigation; the Royal Irrigation Department went on to build the so-called "miracle canal" (เหมืองมหัศจรรย์), a weir and 8-km main canal worth over 30 million baht that eased water shortages for local farmers.





From chemical to organic rice

Mr. Chin Sutha "Po (Father) Chin" left school after grade 3 to farm with his parents, growing oranges, litchi, longan, hill rice and tobacco. He knew that heavy chemical use in the orchards was unhealthy and costly but carried on like his neighbours. A member of the mill since its founding, he holds ten shares. In 2007 the Provincial Land Development Office encouraged him to grow "safe" rice, which he did for two to three years.

> ***"I sell my organic rice to the mill and get a very good price — better than the market price."***
>
> Mr. Chin Sutha, member of the Royal Rice Mill since its founding.

Around 2010–11 (B.E. 2553–54), as he recalls, he attended a training at the mill led by Ajarn “Tik” (Dr Tansiphorn Na Nan), where he first heard about PGS Nan, and joined the Chulalongkorn University programme. His first organic season was a failure: not yet knowing how to control weeds, and unsure he could manage pests, he harvested nothing. Government training at the mill then taught him to control weeds through water management, and yields improved in the following years. By 2013 (B.E. 2556) he was confident enough in the network to convert a second plot, a mixed fruit orchard, to organic.

He also brought his older brother, Mr. Bunchuay Sutha, a former construction contractor whose main income comes from rubber plantations, into PGS Nan; Mr. Bunchuay farms only his paddy fields organically.

Organic Rice under PGS

The School of Agricultural Resources of Chulalongkorn University, through its Nan learning centre, joined provincial meetings on the farming crisis and proposed organic farming certified through a Participatory Guarantee System (PGS), a locally focused, peer-based guarantee built on trust, social

networks and knowledge exchange. Because local farmers knew little about organic farming, the School chose a participatory approach: from 2014, fourth-year undergraduates carried out their senior projects with rice farmers in Tha Wang Pha, documenting their practices, introducing organic methods and designing an organic rice business model based on PGS. Over two years the group was convinced to change its practices, and the idea spread to other districts.

The School also secured a market by partnering with the Lemon Farm Organic PGS network, run by the Health Society Company, which operates organic shops in and around Bangkok. Formal training began in March 2017; in December 2017 the "PGS organic agriculture Nan Group" was established with 35 households in six districts, including Tha Wang Pha, and in March 2018 it was certified by Lemon Farm Organic PGS.

Members of the PGS group guarantee each other's produce: they visit and audit one another's farms, plan production together and carry out quality control, backed by training on organic standards, farm management and certification. As members increasingly learned from each other through peer review and learning by doing, the university's role shifted from teacher to advisor and facilitator. The mill's hall hosts training on organic rice and on water management for weed control, and certified organic paddy is milled and sold by the mill, part of it to Lemon Farm.

TIMELINE SUMMARY

2006
B.E. 2549
Major flood in northern Nan; Royal Rice Fund set up with money, rice and seed.

2007
B.E. 2550
Mr. Chin starts growing "safe" rice with the Provincial Land Development Office.

2009
B.E. 2552
Royal Rice Mill opened by the Princess (27 Feb) and run as a member cooperative.

2010–11
B.E. 2553–54
Royal visit leads to the "miracle canal". Mr. Chin joins PGS Nan with Chulalongkorn.

2014–16
B.E. 2557–59
Chulalongkorn students work with Tha Wang Pha rice farmers; link with Lemon Farm.

2017
B.E. 2560
First PGS training; PGS organic agriculture Nan Group founded with 35 households.

2018
B.E. 2561
Certified by Lemon Farm Organic PGS (Mar); 46 households in 9 districts (Jun).

2019
B.E. 2562
"Krasib-Raks" brand and online sales as supply outgrows Lemon Farm demand.

Activities and Services

A one-stop service centre for rice farmers

In normal times the mill works as an all-round agricultural service centre for its members. Its business includes:

- **Paddy collection** from members and other farmers, at 0.50–1 baht/kg above the local market price depending on quality.
- **Milling and sale of milled rice** to over 30 regular public and private buyers.
- **By-products** (bran, broken rice, husk) and a **milling service** for households across Nan.
- **Farm supplies** (fertiliser, rice seed, general goods), **low-interest credit** for members.
- **Coffee processing** and the Phu Phayak coffee shop and souvenir shop at the mill's learning centre.

The mill processes around 6 tonnes of rice a day in a GMP-certified facility and sells safe and organic rice under the **"Khao Nan"** (ข้าวน่าน) brand, including Riceberry, red jasmine, white jasmine and purple glutinous rice (khao niao kam). Its roadside location makes it a popular stop for visitors exploring Tha Wang Pha.



Learning with young people

Since 2014 the mill has run youth training on rice and mill management. In May 2017, 200 secondary students from Tha Wang Pha Phitthayakhom School followed the whole chain from rice cultivation to the work of the mill, then applied the cooperative model to their own school.

Business Model and Networking

A cooperative enterprise

The Royal Rice Mill Group is owned and run by local residents following cooperative principles. Shares cost 10 baht each, with a minimum of ten on joining and no upper limit. Members are organised in 13 groups — one for each village of Si Phum (three of them in Ban Don Tan) and one for neighbouring sub-districts. In 2016–17 the mill had 1,196 members, a 15-member board drawn from village leaders, two auditors, a management team of four and three hired staff. Salaries and running costs were covered entirely by the mill's own income, without further government support, and it paid dividends every year since its founding without recording a loss.

Net profit is allocated each year by the general assembly: at least 10% to the reserve fund, at least 20% to the Royal Rice Fund's disaster reserve, at least 10% to a development fund, at least 5% each to member welfare and to public benefit, and the remainder to dividends on shares and to patronage refunds based on the paddy members sell. The coffee shop has become a significant earner: its revenue rose from 0.44 to 1.52 million baht between FY2014 and FY2015. In FY2016 the mill collected 359 tonnes of paddy (target 400) and sold 151 tonnes of milled rice (target 200), reflecting strict quality control on purchases.



Partnerships, buyers and support

As a royal project, the mill has enjoyed smooth relations with agencies at provincial, district and sub-district levels and with the private sector, bringing funding, knowledge and markets with little conflict. Key partners include the Office of HRH Princess Sirindhorn, the Nan Provincial Cooperative Office, Si Phum SAO, Nan PAO, the Royal Irrigation Department, the Provincial Land Development Office and the Rice Research Centre.

Current challenges

An organic market bottleneck. As the PGS group grew from 35 to 110 households in two years, supply quickly outran Lemon Farm's demand. Because Lemon Farm sells to small retail customers, it buys in small lots: organic rice built up in the mill's stock, which then could not take in new harvests, putting pressure on the mill. Some members have stepped back. Then, the organic rice market was hit hard during and after COVID as consumers could not afford high price premiums. Many members withdrew because organic volumes did not match purchases and switched to "safe standard" coloured rice with a wider market.

Ageing farmers and succession. Mrs. Jurirat Intrawang "Mother Jui" came back from 20 years in Bangkok when her father fell ill; she grows organic rice but her children are not interested in farming, so she will continue "as long as her body allows". The mill faces the same question: young people leave for the city, capable villagers decline board seats, members tend to re-elect the same village leaders, and the sense of ownership is weaker than in mills built solely from villagers' own savings.

Reccuring disasters have also dramatically constrained rice production in the area. Mr. Chin and his neighbours were hit by heavy floods and did not get any rice harvest three years in a row in 2024, 2025 and 2026 which dramatically reduced the volumes of organic rice treated by the Royal Mill, only 600 kg in 2025, which put the initial business model at risk.





Wat Pong Kham Learning Center

Community Learning Promotion Foundation · Du Phong, Santisuk District, Nan

Forest, farm and Dhamma — Buddhist agriculture for the well-being of farmers and forests

Background

Travelling across Nan Province offers sweeping views of rolling mountains, and that is the problem: few trees now block the view. Nan had some of the worst deforestation in Thailand. Its forests declined by about a third between 2000 and 2012, even as forest cover was recovering elsewhere in the country. Around 80% of the province is classified as protected land, yet only about 14% of its people hold title to the land they farm.

Without secure tenure, farmers plant maize on hillsides: it grows fast, needs no long-term investment and sells well. Companies supply seed and agrochemicals on credit, repaid at harvest. When crops fall short of company standards, farmers fall into debt and clear more forest. Bare slopes bring erosion, chemicals in the streams, worse floods and harsher droughts. Poverty, household debt and labour migration follow.

Wat Pong Kham stands in Du Phong sub-district, Santisuk district. Its abbot, Phrakhru Sujinnanthakit (Somkit Jaranathammo), traces deforestation and debt to greed and delusion and answers with what he calls **dhammic agriculture**: farming on a Dhamma basis that joins Buddhist ethics to scientific knowledge and market strategy. Since 1992 the temple has been a community learning centre, and since 2008 a registered foundation at the hub of a network of organic community enterprises.

NAN AT A GLANCE

~33%	~80%	~14%	~50
decline in Nan's forests, 2000–2012	of the province classified as protected land	of Nan's people hold title to their land	farmers in Somkit's cooperative (2019)





Origins of the Learning Center

A monk from a farming family

Phrakhru Somkit grew up in Pong Kham and was ordained as a novice at 12. He saw his own family's livelihood change. In the early 1980s, after the communist insurgency ended, his father was among the first in the village to give up upland subsistence farming for maize. As for many families, prosperity did not follow: crops often failed company standards, debts mounted, and more forest was cleared.

Learning from other monks

In the early 1990s, Somkit joined Phrakhru Pithak Nanthakhun of nearby Ban Kiw Mueang, his teacher and some fifteen years his senior. Pithak performed Nan's first tree ordination and founded the Hug Mueang Nan network. Somkit was also inspired by Phrakhru Manas of Phayao, pioneer of tree ordination. Alongside his monastic career he earned a bachelor's degree and a master's in Man and Environment Management at Chiang Mai University (2005), a scientific grounding that helps him work with NGOs and state agencies.

Three stages of activism

Somkit describes his work in three stages. The first, beginning in the 1980s and 1990s, aimed to preserve the forest. The second in the 2000s fought debt through village savings funds: "Still, we couldn't stop the debt." The third in the 2010s is dhammic agriculture, which goes beyond subsistence: farmers should "grow less but gain more", focusing on quality so that consumers want their produce.

A drop of dye in clear water

From 1997 to 2015 (B.E. 2540–2558), Somkit watched with growing unease as maize spread across Nan. He also saw monks' standing decline, as development came to mean money, degrees and official endorsement, leaving monks only to look after the temple as a "bridge of merit". Yet the Buddha called the Sangha the world's field of merit, meant to relieve physical and mental suffering. In the name of development, he argues, farming was pulled away from the Dhamma and reduced to "whoever does more gets more", forgetting that doing more also destroys more. His answer was to start small, where the problem lies, and let the example spread.

Farming is unusual for a monk, as the monastic rules forbid digging the soil. Somkit sees caring for land as part of relieving suffering, while remaining a "field of merit" through which villagers make merit.

> ***"We might ask to be a dye, a primary colour dripped into water. Even if it is only a tiny drop, it will gradually spread and may turn clear water into coloured water."***
>
> *Phrakhru Somkit, on starting small*

A hat for the mountain, shoes for its foot

The project "Putting a Hat on the Mountain, Putting Shoes on the Foot of the Hill" (Suam muak hai phukhao, suam rongthao hai tin doi), i.e. restoring forest cover on the hilltops (the "hat") while farming the lower slopes and foothills (the "shoes") grew from lessons learned on forest walks: start from a small problem spot, then expand.

Around 1991 Somkit adapted the traditional Phra Phaa offering: instead of goods, he asked villagers to give trees, and replanted the maize fields behind the temple, turning it into a forest temple (wat pa). In the mid-1990s his father gave him a worn-out maize plot as an act of merit. Left fallow for about ten years, it became lush forest again, rich in plants and animals. People said the monk had gone on bindabat, the morning alms round, for land, and other villagers began to donate land too.

WAT PONG KHAM AT A GLANCE

1992 community learning centre set up at the temple	**2008** Community Learning Promotion Foundation registered	**29** communities in the temple's network	**3** temples following the Wat Pong Kham model

The Temple, Forest, and Farm

Rituals for forest and river

Eco-Buddhism is woven with ritual: tree ordinations (buat pa), life-prolonging ceremonies for rivers (suep chata mae nam), and "alms-requesting" of forest, in which land is asked back from farming and returned to forest.

Somkit links Buddhist teaching with local belief: "good soil is protected by a beneficent spirit, Mae Thoranee, goddess of earth … Good water is protected by the water goddess, Phramae Khong Kha." People honour these spirits and care for what they protect. For Somkit the forest "is essentially a part of life", providing water, soil, food and knowledge.

THE FOUR HOUSEHOLDER VIRTUES

Dhammic agriculture applies the four values the Buddha taught for lay householders (gharavasa-dhamma): honesty, tolerance, patience and sharing. Through them, Somkit argues, farmers can move from capitalist to dhammic agriculture.

Integrated farming

Visitors may think they are still walking through woods until fruit trees wrapped in pea vines come into view. Integrated agriculture (kaset phasom phasaan) mimics nature: plants that help each other grow together, fish live in ponds, and chickens roam freely, eating insects and fallen fruit. The farm shows intensive planting on a small plot, with trellises around fishponds whose water and plants nourish the crops.

Member farmers have moved from maize to mixed organic vegetables grown in greenhouses with drip irrigation and compost: Chinese cabbage, choy sum, baby bok choy, Japanese cucumber, tomato, pea aubergine, pumpkin, passion fruit, ginger, galangal and lemongrass. Low costs, a secure buyer and a steady income are what make the system work for them.

Marketing on a Dhamma basis

Farmers always ask: who will buy it? Somkit uses faith as a marketing lever. Giving talks in other provinces, he took along a tonne or two of unsold produce: "People without merit will not get the chance to eat this produce." It sold out. The foundation now collects members' vegetables, sold through retailers as "morality vegetables of Wat Pong Kham" under the slogan "Love Nan, love the forest, love water, love life".

A foundation and an enterprise network

The Community Learning Promotion Foundation (Wat Pong Kham) works on three principles: farming based on community needs, business on a Dhamma basis, and a network of community enterprises. Because the monk's group was dismissed as unofficial, members registered as community enterprises and joined a participatory (PGS) organic network. Somkit became its chair, the first monk in Thailand to chair a community enterprise. Shared rules stress honesty, patience, self-reliance, sacrifice, sharing and, above all, action.

A Sufficiency Economy model

The temple also applies the Sufficiency Economy model of King Rama IX: moderation, reasonableness and self-immunity, built on knowledge and virtue, adapted to local conditions. As a learning centre, the temple works in four dimensions:

1. **Economic** — upland crops, small intensive plots instead of maize, ponds and water systems, greenhouses, processing, local seed saving
2. **Social** — cooperation between home, temple and school, farmer-group development, networks inside and outside the village
3. **Environmental** — restoration and conservation, and a network to monitor and report local problems
4. **Cultural** — village festivals, a community-history curriculum, and the temple as a cultural learning site

TURNING CAPITAL INTO DHAMMA

Dhamma as the base: nature, the Buddha's teaching and local ways of life.

Work as capital: grouping, joint activities, networks and shared planning.

Merit as the goal: diligence, honesty, generosity and love of one's homeland.





Activities and Training Content

Partners and the royal network

The temple works with many partners from farming communities, civil society organizations, private sector and government agencies. A Royal Project supplies greenhouse plastic and seeds and buys produce. Somkit has promoted vetiver grass against erosion, advised the Phu Fa royal project in Bo Kluea, and since 2009 has served as an adviser to the Phid Thong Lang Phra royal project for the upper Nan basin.

Current challenges

Uptake in the home village (Pong Kham) remains limited: only about 30 of 240 households belong to the organic enterprise. Many villagers understand the Sufficiency Economy but still seek cash through migrant work and monoculture, and cite lack of capital, costly fertiliser, unstable prices, too little land and water shortages. Researchers also debate the model.

Six training courses

1. **Seeds and seedlings** — raising seeds and collecting saplings
2. **Life planning** — changing mindsets, saving money and "saving life"
3. **Soil fertility** — quality organic fertiliser and bio-ferments
4. **Processing** — transforming and packaging farm produce
5. **Planning and marketing** — production plans and market access
6. **Farm management** — crop plots, livestock and fish

TIMELINE SUMMARY

Early 1980s
B.E. 2520s

Somkit's father is among the first in the village to grow maize.

Early 1990s
B.E. 2530s

Somkit joins Phrakhru Pithak; farming begin behind the temple.

1992
B.E. 2535

Wat Pong Kham becomes a community learning centre.

Mid-1990s
B.E. 2530s

Father donates a worn-out maize plot; left fallow, it returns to forest.

1997–2015
B.E. 2540–2558

Maize expands across Nan; Somkit seeks a way out.

2005
B.E. 2548

Completes his master's thesis at Chiang Mai University.

2008
B.E. 2551

Community Learning Promotion Foundation registered.

2009
B.E. 2552

Somkit advises the Phid Thong Lang Phra royal project.

2016
B.E. 2559

"Hat on the Mountain" project; enterprise network formed.

2019
B.E. 2562

About 50 farmers in the cooperative; produce sold in Bangkok.

2020
B.E. 2563

Forum on organic standards, greenhouses and marketing.

Now
Ongoing

Seeking Thai FDA approval for herbal products development.











ORIGI NAN

Nan Processed Agriculture Cluster · Mueang Nan · Nan Province

Many producers, one local identity — adding value to Nan's wisdom

Background

Nan is a mountainous headwater province in Thailand's upper north. Its villages hold a wealth of local raw materials: *mafai jin* (Chinese wampee), *mak khwaen* (Sichuan pepper), golden oranges, peanuts, honey, herbs, highland coffee — and the know-how to process them. Yet small community enterprises working alone struggle with the same barriers: brands too small to be noticed, costly marketing and logistics, and standards that are hard to meet.

OrigiNan, the Origin of Nan, grew from a simple conviction shared by local processors: **a business on its own cannot go as far as a network**. Many of its members belong to a new generation of entrepreneurs who chose to return home and give Nan's resources creative value. OrigiNan is the **processed agriculture cluster** of Nan Province — 66 member communities and groups in 10 of the province's 15 districts — and the **umbrella brand** under which members develop and sell products made only from Nan raw materials. Its home is the **Origi-nan Space** at Ban Thua Lisong, near Nan airport.

The cluster is one of seven economic clusters recognised by Nan Province (tourism, community spirits, herbal medicine, processed agriculture, handwoven textiles, silverware and large-scale farming). Its work gained new resonance in 2025, when Nan was designated a **UNESCO Creative City of Crafts and Folk Art**.

Origins of OrigiNan

From processors to a cluster

In 2017 (B.E. 2560) the Ministry of Industry was promoting business clusters across Thailand. Ms. Aree Phetrat of Ban Thua Lisong, a family peanut-processing business, attended the training and saw its value — but Nan then had clusters only for silverware and handwoven textiles, none for agriculture. Talking with fellow entrepreneurs, among them Mr. Thirayut Chankhlai, a processor of Chinese wampee and other local fruit products, they found their common ground in the midstream: processing. Seven or eight communities founded the processed agriculture cluster.

The same year it was legally registered and certified by the Nan Provincial Industry Office, with recognition from the Department of Industrial Promotion, Region 1 (Chiang Mai). Development funding followed for about five years (2018–2023): used to build the cluster's roles and working processes. Membership rose from 20 in 2017 to 77 in 2018–2020.

COVID and the turn to markets

COVID-19 cut public budgets and stopped travel — and sales. Before the pandemic the cluster had concentrated on product development; afterwards it understood that **online markets, storytelling, content and digital tools** were essential. When funding resumed in 2022, the cluster increasingly raised its own resources through pitching competitions and university partnerships.

A shared brand

To give members' diverse products a common face, the cluster created the OrigiNan brand: products must be made from Nan raw materials and carry the province's identity. Its signature ingredient is Nan's GI-registered golden orange (*som si thong*), worked into essential oil, jam, cookies, liqueur and cosmetics. When a Thai Senate delegation visited in 2023, the cluster was cited as a model of regional cluster development, and in 2024 OrigiNan received an Outstanding Innovation Award.

TIMELINE SUMMARY

2017	2018–20	2020–22	2023	2024	2025
Cluster founded by 7 community enterprises and certified.	Public development funding; membership peaks at 77.	COVID: budgets cut; shift to online markets and content.	OrigiNan brand built around the GI golden orange.	Origi-nan Space opens; Outstanding Innovation Award.	New vision and missions; Nan named UNESCO Creative City.

How the Cluster Works

Five missions

1. Carry Nan's local wisdom forward with **technology and innovation** to create added value.
2. Promote the **creative economy and BCG** (Bio-Circular-Green) principles among members.
3. Run a **mutually supportive business** sharing knowledge and raw materials rather than competing.
4. Foster **new businesses and grow existing ones**: the OrigiNan brand and members' own brands coexist.
5. Become a **business academy and service centre**, earning income from members' workshops, not just products.

Governance on a shoestring

Mr. Thirayut Chankhlai chairs the cluster; Ms. Aree Phetrat is its Cluster Development Agent (CDA), who carries the vision across changes of leadership. The cluster has no full-time staff: members take turns handling meetings, visits and interviews for a small allowance (200 baht per task). Members pay a 700-baht annual fee, and 3% of OrigiNan-brand sales goes to a central fund for management costs. The Provincial Commerce Office acts as the cluster's government mentor.

Growing members step by step

New members first register their products, raw materials, standards and production capacity in the cluster database. They then enter a learning process: rather than marketing for them, the cluster teaches members to sell — storytelling, content, packaging design, pricing, offline and online marketing. Products that are ready in packaging, price and volume go straight to market, and their success motivates others.

A COMMON VISION

To elevate Nan's local wisdom together with fellow entrepreneurs through the creative economy, and to be a model of mutually supportive business for the pride of the people of Nan.

Members, Products and Markets

66	390	240 t+	1,500+
member communities & groups	people, in 10 of 15 districts	Nan raw materials used per year	visitors & workshop learners

1 · Agricultural produce
raw materials & processed crops: peanuts, Chinese wampee, mak khwaen, oranges

2 · Local wisdom & culture
crafts and products rooted in Nan's heritage, e.g. natural tie-dye

ORIGI Nan

3 · Farm & tourism products
honey, salt and other farm goods; e.g. Luk Chan PGS Nan

4 · Health & well-being
herbal inhalers, shampoos, Nan herbs; e.g. Bio-way, Baan Nam Kian

OrigiNan's four member categories, with examples

Developing the OrigiNan brand

Each year, members pitch new OrigiNan products; only members with the required standards and experience are selected, and grants cover the development period only, afterwards the maker produces at their own cost. Funding has come from the Nan Provincial Administrative Organization (year 1), an NIA pitching programme (year 2) and the Regional Industrial Promotion Center 1 (year 3). Universities fund specific lines: orange jam and other foods with RMUTL Nan; shampoo, lip balm and face foam with Uttaradit Rajabhat University. The 2024 line added grain products, orange cookies, orange liqueur and banana chips with orange jam. Seasonal limited editions showcase Nan's diversity, such as the Nan Honey box set of seven wild honeys from seven botanical sources across the province.

Partners

The cluster gathers members' needs and then brings in the agency best placed to meet them. Partners include the Nan Provincial Industry, Agriculture and Commerce Offices; the Nan PAO; NIA; universities (RMUTL Nan, Uttaradit Rajabhat) and science parks in Phayao, Uttaradit, Chiang Mai and Chiang Rai. Cooperation depends on each agency's budget and objectives.

Shared markets and logistics

OrigiNan sells B2B and B2C, online and offline, across Thailand, including through the Chaipattana Foundation shop, with export as a long-term goal. Ban Thua Lisong runs the community service centre: a **live-streaming room**, a **community warehouse** and **storefronts** on Shopee, Lazada and LINE OA. Members can sell in the shop or online; when an order comes in, Ban Thua Lisong **packs and ships** it so that members can focus on production. Members also share raw materials with one another.

A local business academy

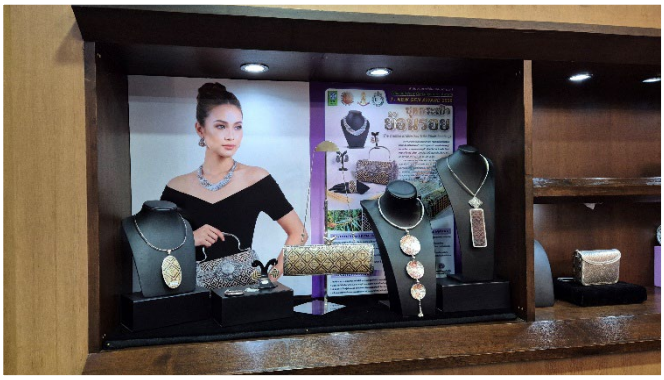
Workshops are a growing income stream for members, who design and teach them. Each workshop lasts about 40 minutes, and every participant takes home what they made. Schools choose from the list of members-proposed workshops according to their goals. More than 1,500 people have visited or trained at OrigiNan — students, and also groups such as 100-plus housewives from Wang Thonglang in Bangkok, who went home able to make and sell products themselves.

Opened in June 2024 beside Ban Thua Lisong on Route 101, the Space combines a gallery telling the cluster's story, a Nan coffee and peanut-milk bar, a craft corner, and meeting rooms for 60 and 10 people.





Promoting a large diversity of members' activities

















Origi-nan Space, Route 101, near Nan Nakhon Airport, 113 Moo 2, Pha Sing Sub-district, Mueang Nan District, Nan Province, Thailand

Hours Daily · shop 08:30–17:30 · Origi-nan Space 09:30–16:30

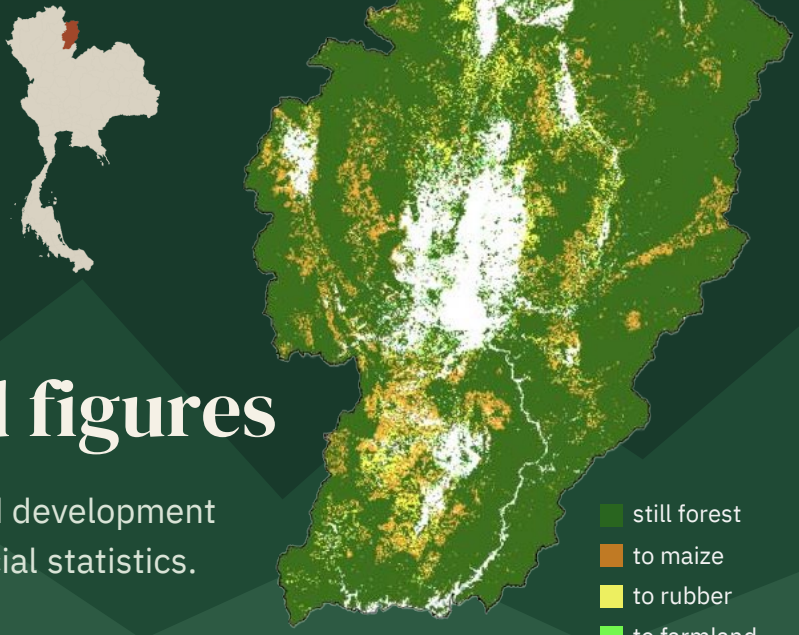
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OrigiNan on Facebook

FACT BROCHURE · OCTOBER 2026

Nan

THAILAND



Kruasilp et al. 2023 [21]

Province facts and figures

Agriculture, forests, people, economy and development plans of Nan province compiled from official statistics.

11,472 km²
land area (≈7.17 million rai), 87% of it mountains

470,507
registered inhabitants, 31 Dec 2024

15 · 99 · 896
districts · sub-districts · villages

61.0%
forest cover, 2025

≈40%
of Chao Phraya runoff comes from the Nan River basin

59.5%
of employed people work in agriculture (Q3 2024)

5
ethnic groups: Tai Lue, Hmong, Yao (Mien), Htin and Mlabri ("Tong Luang").

2.6
people per household on average in 2025 Household Socio-Economic Survey [22].

8.2 yrs
average schooling of people 15+ (2024), 12th of 17 northern provinces

271
doctors and 1,405 registered nurses (2024); 16 hospitals, 147 sub-district health centres

Sources: NSO Nan 2025 [1][2]; RFD 2025 [12]

Mountains, forest and a river head

High ranges in the north, where the Nan River and its tributaries rise, open onto the Nan valley towards the south. Two thirds of the land is classified as forest.

15 districts at a glance, 2024

District	Population	km²	/km²
Mueang Nan	81,528	929	88
Wiang Sa	68,142	1,895	36
Pua	63,717	657	97
Tha Wang Pha	49,847	702	71
Phu Phiang	35,701	392	91
Na Noi	31,681	1,408	23
Chiang Klang	26,514	277	96
Thung Chang	18,893	761	25
Mae Charim	16,422	998	16
Santi Suk	15,471	417	37
Bo Kluea	15,395	848	18
Na Muen	13,854	786	18
Song Khwae	12,392	544	23
Ban Luang	11,180	338	33
Chaloem Phra Kiat	9,770	519	19
Province	**470,507**	**11,472**	**41**

Land use, 2023

Share of 7.17 million rai



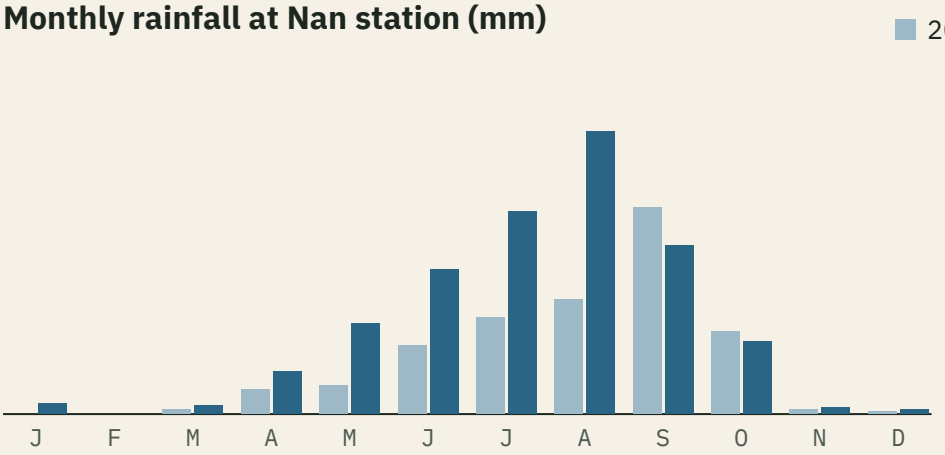
Forest land	**64.6%**
Agricultural land	**20.4%**
Other / unclassified	**15.0%**

Agricultural land (1.47 M rai) by use

Upland field crops	**43.9%**
Fruit & perennial trees	**30.0%**
Paddy	**23.0%**
Other	**2.5%**
Vegetables & flowers	**0.5%**

OAE land-use series. The Royal Forest Department's 2025 satellite estimate, on a 7.58 M rai base, gives 61.0% forest.

Monthly rainfall at Nan station (mm)



471 mm
fell in August 2024 alone, the wettest month of the two years

2024 temperatures: mean 27.6 °C; absolute max 42.0 °C (April); absolute min 13.8 °C

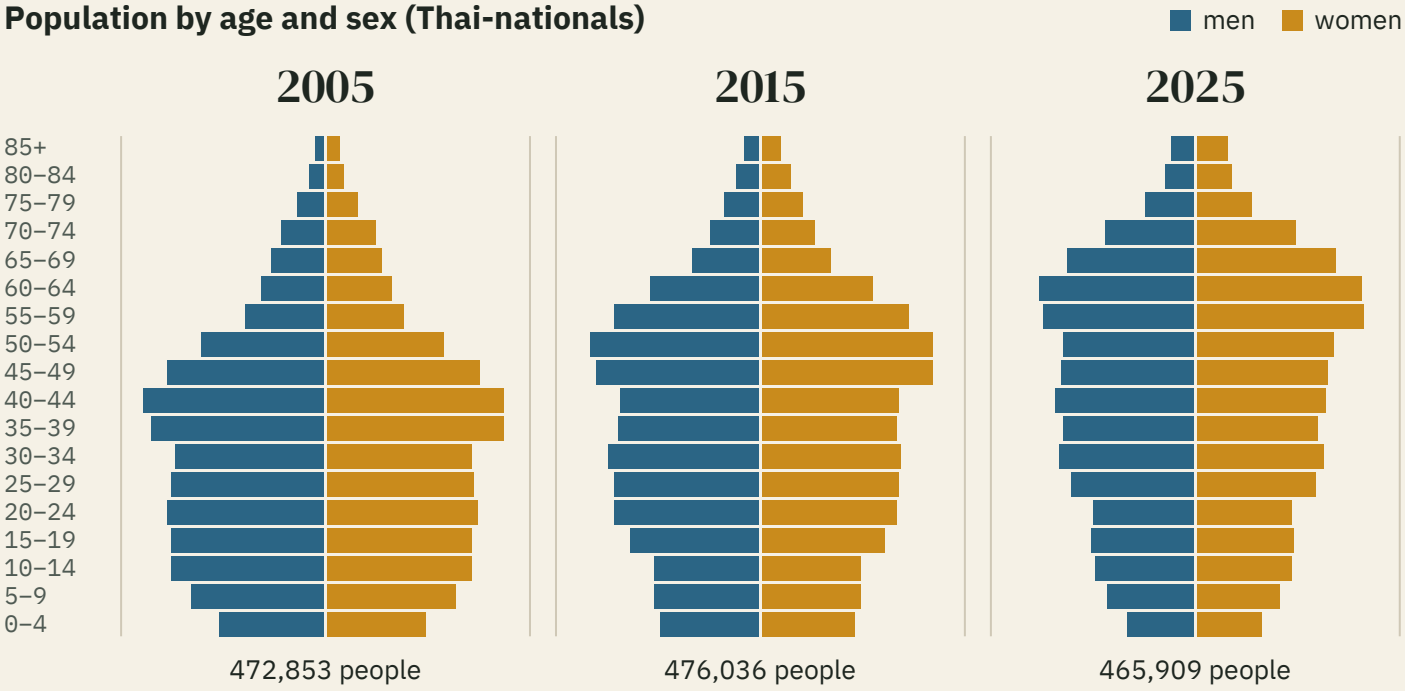
Hot season Mar–Jun · rains Jul–Oct · cool season Dec–Feb

Sources: districts — NSO Nan, Provincial Statistical Report 2025, tables 1.1 & 1.4 (DOPA registration, 31 Dec 2024) [1]; land use — ibid., table 11.1 (OAE) [1]; climate — ibid., tables 20.6 & 20.8 (Nan Meteorological Station) and provincial overview [1]. Density in persons per km².

A shrinking, ageing population

The large generation aged 35–44 in 2005 is now 55–64. Meanwhile births have collapsed: Nan registered 46% fewer infants under one year in 2025 than in 2005.

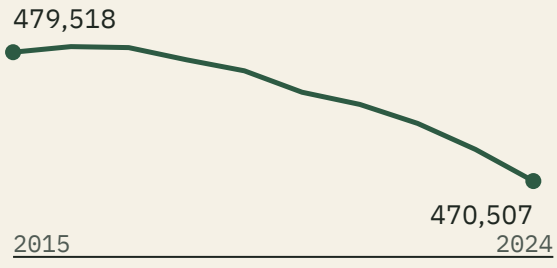
Population by age and sex (Thai-nationals)



Key indicators

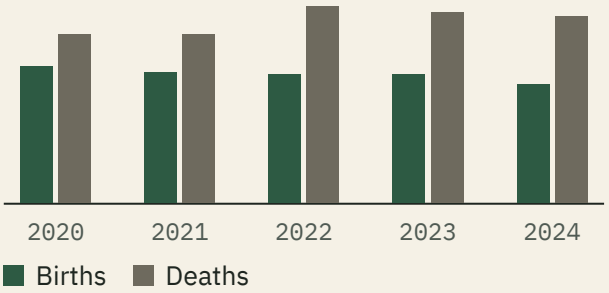
	2005	2015	2025
Share aged 60 and over	11.3%	16.0%	**26.4%**
Share aged 80 and over	1.3%	2.2%	**3.1%**
Share aged under 15	20.1%	15.6%	**13.2%**
Median age (years)	33	38	**44**
People 60+ per 100 aged 15–59	16	23	**44**
Infants under 1 year	4,859	4,197	**2,619**

Registered population, 2015–2024



−9,011 people (−1.9%) in nine years; −0.47% in 2024 alone.

Births and deaths



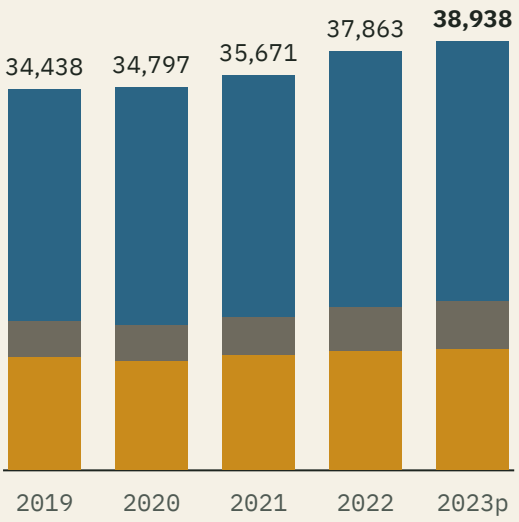
2024: 2,631 births, 4,199 deaths — crude birth rate 5.6‰, death rate 8.9‰.

Sources: Department of Provincial Administration, Nan, December 2005, 2015, 2025 [24] — NSO HSES 2025 [22]; Provincial Statistical Report 2025; Nan Provincial Health Office) [1];

A farm-and-services economy

Agriculture produces more than a quarter of provincial output and employs most workers. Household incomes, below northern and national averages in 2023, rose strongly in 2025. Poverty has fallen sharply over the decade.

Gross provincial product (million-baht, current prices)



89,515 THB

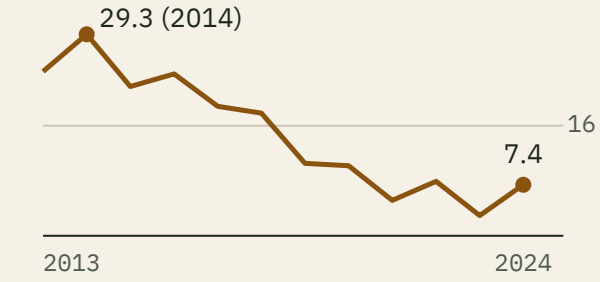
GPP per head, 2023 — 62nd of 77 provinces

Average household income (baht/month)



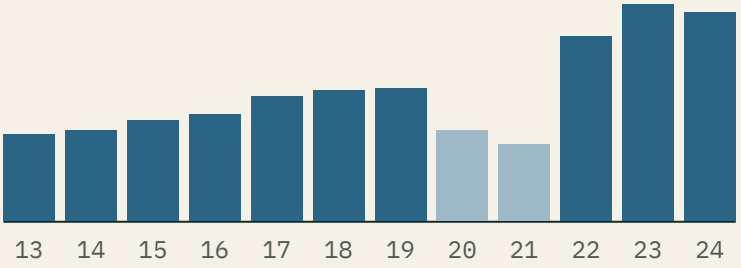
2025 regional and national figures not yet compared. Household finances in detail: p. 6.

Share of people below the poverty line (%)



32,100 poor people in 2024 (consumption-based measure).

Visitors per year (Thai and foreign)



1.51 M visitors in 2024, 1.5% foreign

4,407 M THB visitor spending, 2024

70.8% hotel occupancy, 2nd in the North

199

registered factories (2024), 3,250 M baht invested, 1,962 workers; 0.27% of Thailand's factories

Labour force, Q3 2025

0.3% unemployment rate



243,397 employed **1,644** unemployed

123,674 outside the labour force

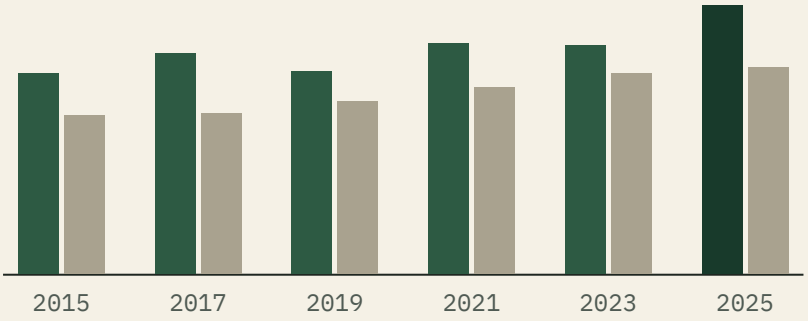
Of 368,715 people aged 15+. Average monthly wage in Q3 2024: 13,805 baht.

Sources: GPP — NESDC via NSO Nan Report 2025 [1]; income 2025 — NSO HSES 2025 [22]; income 2023, poverty, factories, visitors (Ministry of Tourism & Sports, 2013–2024 series) — Nan Pocket Book 2025 [2]; foreign share — Report 2025 [1]; Bars for 2020–21 (pale) = COVID-19 years.

Highest incomes in a decade, debt easing

The 2025 Household Socio-Economic Survey shows falling debt and lower inequality — yet a quarter of income still comes from transfers, not work.

Monthly income and spending per household (THB)



income: 17,886 → **23,904** spending: 14,165 → **18,355**

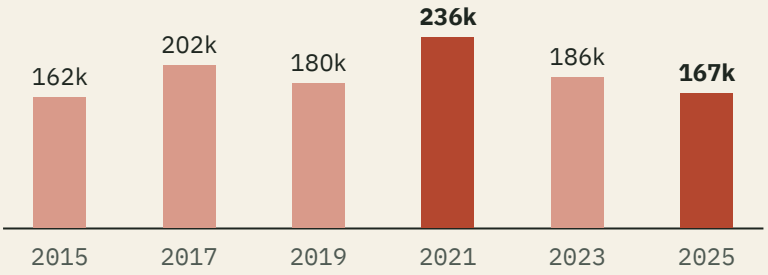
Where 2025 income comes from



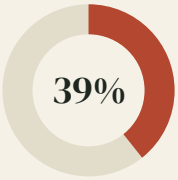
Wages & salaries	**5,471**
Farm profit	**4,561**
Business profit	**2,758**
Transfers & aid	**6,059**
Income in kind	**4,351**
Property & irregular	**704**

Work brings 53.5% of income; farming alone 19%.

Average debt per household (baht)



Averaged over all households. Debt equals 7.0 months of income in 2025, down from 9.2 in 2023.



of households in debt in 2025, down from 46.5% in 2023. Almost all debt is formal; the BAAC is the main lender for 48.7% of indebted households.

What the debt is for (2025)

Consumption	44.5%
Farming	21.5%
House or land	16.9%
Business	15.1%

0.266
Gini coefficient of household income, 2025 (0.305 in 2023)

7,314 THB
monthly food, drink & tobacco spending — the largest item

5,549 THB
average monthly surplus of income over spending

Enterprises — 2022 Business & Industrial Census

25,375 active establishments (2021)

Manufacturing	38.9%
Retail trade	25.4%
Other services	9.0%
Food & drink services	8.7%
All other activities	18.0%

99.3%
have 1–15 workers; only 16 have more than 50

95.3%
are owned by individuals

50,371
people working in them

9,552
of 34,927 units listed had closed, paused or moved

Sources: NSO, Household Socio-Economic Survey 2025 — Nan Province (published 2026) [22]; NSO Nan, 2022 Business and Industrial Census, Nan [23].

107,796 farm holdings

The 2023 Agricultural Census portrays small, mostly crop-based farms, widely mechanised, heavily indebted and exposed to rising input costs and drought.

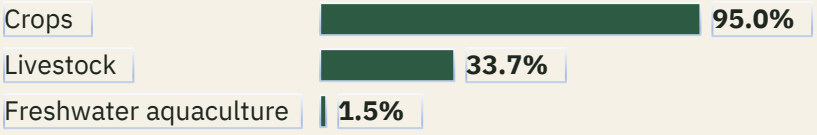
1.76 M rai
farmed by holders (≈282,000 ha)

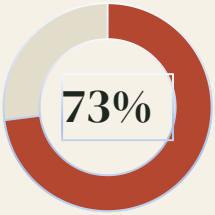
16.3 rai
average holding (≈2.6 ha)

98.2%
of holders on the farmer register

35.4%
use farming apps (to register with government, check weather)

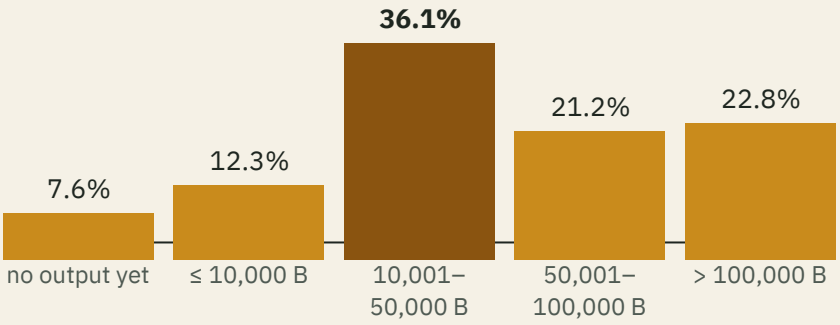
What holders do (several answers possible)

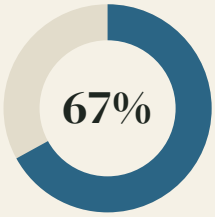




of holders are in debt (63,066). Average debt per farm household: **209,359 B**

Annual farm income per holding

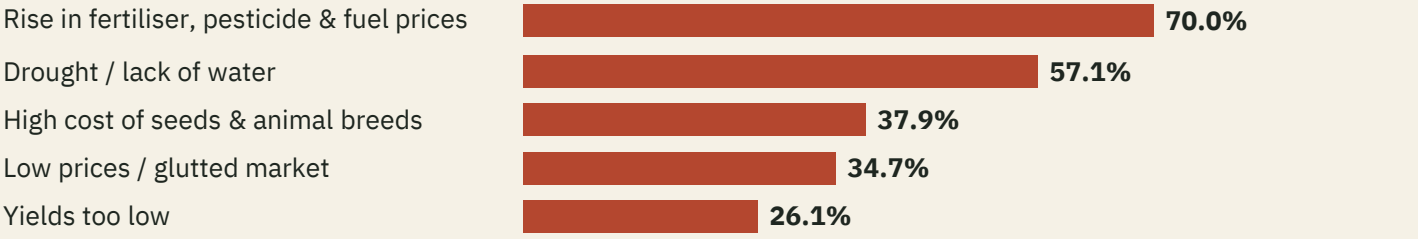




use machinery (54,463). Most common: 4-wheel tractor (54.9% of users), hand sprayer (46.9%), hand weeder (29.3%).

Top five problems reported by farm holders

74.3% reported at least one problem or disaster



What villages ask for, 2024 survey

52.5% solutions to farming problems

28.1% household water supply

25.6% land title documents

19.3% road repair

18.4% debt relief

Sources: NSO, Agricultural Census 2023 — Nan results (infographic) [3], reproduced in Nan Pocket Book 2025, p. 12 [2]; village needs — NSO survey of village problems and needs 2024, Pocket Book p. 10 [2]. 1 ha = 6.25 rai.

Maize, rice and rubber cover the land

Three crops account for most of the planted area. Rice of all kinds — lowland, dry-season and upland — covers about as much land as feed maize.

Planted area, crop year 2024 (rai)

field crops · rice · tree & fruit crops

Crop	Area	Output (t)
Feed maize	484,052	329,644
Lowland rice (wet season)	332,059	166,483
Para rubber	329,606	47,675
Upland rice	121,309	46,343
Cassava	78,188	249,241
Longan	60,774	27,595
Dry-season rice	26,876	15,496
Mango	18,618	8,328
Coffee	12,945	7,795
Lychee	12,196	2,415
Rambutan	9,970	3,514
Ginger	3,415	10,311
Tangerine	3,196	3,092
Oil palm	3,048	1,914

480,244 rai

of rice in all forms; glutinous (sticky) rice makes up 57% of wet-season lowland rice

GI

"Som Si Thong" golden orange holds a geographical indication

332,520 rai

of rubber standing in 2025, 274,346 rai tapped (OAE–DOAE agreed figures); 34,203 growers have mapped 355,478 rai of registered plots (July 2026)

+13%

coffee area 2022→2024 (11,484 → 12,945 rai), listed with sericulture and herbs among Nan's high-value products

Livestock, 2024 · 60,053 livestock farmers

1.92 M chickens

77,414 pigs

68,218 ducks

63,689 beef cattle

10,139 buffaloes

3,454 goats & sheep

Inland fisheries, 2024

2,926 t

from 11,241 fish farms (ponds, rice-fish, cages)

1,498 t

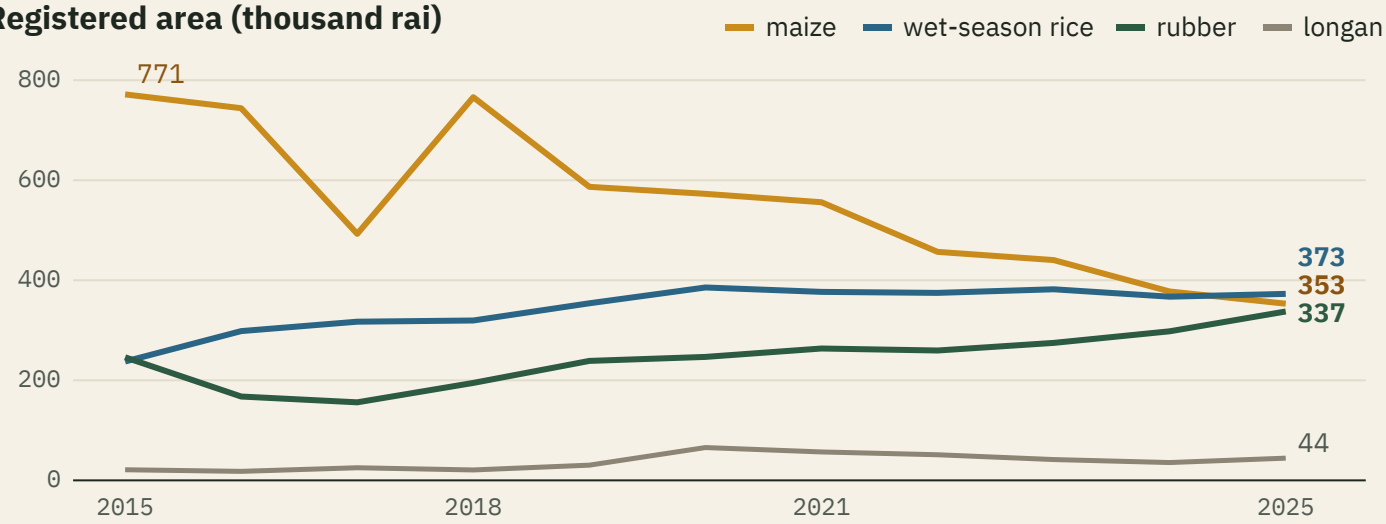
wild freshwater catch

Sources: NSO Nan, Provincial Statistical Report 2025 (Nan Provincial Agricultural Extension, Livestock and Fishery Offices) [1]; coffee 2022 — Report 2023 [4]. Figures come from farmer registration and may not be exhaustive (Nan Provincial Agricultural Office, 2026). High-value products: Pocket Book [2]. Rubber 2025: OAE–DOAE data-unification tables [26]; plot registration GIS summary, 15 Jul 2026 [27].

A decade of crop switching

Farmer-registration records show registered maize land more than halving between 2015 and 2025, while rubber and rice grew. Registration follows support schemes, so shifts partly reflect who registers, not only what is planted.

Registered area (thousand rai)



−54% maize area; growers 34,036 → 20,167

+57% wet-season rice area

+37% rubber area; growers 22,391 → 24,474

×2.1 longan area

Registered maize by district (thousand rai)

District	pale 2015 · dark 2025	change
Wiang Sa		−32%
Na Noi		−41%
Mae Charim		−72%
Tha Wang Pha		**−93%**
Mueang Nan		−66%
Na Muen		−77%
Santi Suk		−79%
Phu Phiang		−86%
Pua		−73%
Chaloem Phra Kiat		−52%
Song Khwae		−77%
Ban Luang		−29%
Bo Kluea · Chiang Klang · Thung Chang	each under 7,000 rai in both years	

Wiang Sa alone held 48% of registered maize in 2025 (170,924 rai).

Farm support at a glance, 2024

63,840 rai
irrigated area; 83 small and medium reservoirs hold 68.7 million m³

137 groups
"large-plot" collective farming schemes, 28 commodities, 9,478 farmers

7,387 farmers
GAP-certified for crops (34,178 rai); 230 farmers organic under PGS (1,471 rai)

1,599 · 517
Smart Farmers · Young Smart Farmers; 15 main and 176 network farmer learning centres

Sources: DOAE farmer-registration data, Nan, 2015–2025 [25]; farm support — Nan Provincial Agriculture and Cooperatives Office, "Agricultural situation and development directions, Nan 2024" [29].

The maize boom and the forest

Feed maize spread quickly up Nan's slopes in the 2000s, driving forest loss. Its area has been shrinking in recent years, while the province experiments with new land-use arrangements.

Seen from space, 1990 → 2019

Satellite classification, km²; panels use different scales



Para rubber grew 6.5-fold between 2007 and 2019; satellite maize area fell 17.6% in 2014–2019. In all, 1,298 km² of forest changed cover over the 30 years (map on the cover) [21].

Official maize-area figures differ by agency (e.g. 2023: 504,728 rai in NSO tables, 564,562 rai by OAE Region 2 [1][20]). The Commerce Office estimated 372,090 rai for 2025/26 [5].

≈1.2 M rai

of forest lost in Nan between 1964 and 2018 [9]

Average forest loss per year (rai)



66,072 ha

of forest lost 2001–2016 (9.1% of the 2000 cover); about 92% became cropland, mainly maize. Annual loss peaked in 2012 (7,523 ha), the year world maize prices peaked [8].

Forest area today

4.63 M rai

forest in 2025 = 61.0% of the province [12]

−10,385 rai

change 2024→2025 (−0.22%) [12]

Land legally designated as forest (6.50 M rai)



2.78 M rai conservation forest

3.71 M rai 17 national reserved forests

Much farmland and many settlements lie inside these legal forest zones; resolving the resulting disputes is a stated aim of the Nan Sandbox [10]. In 2024 GISTDA mapped 2.70 M rai (35.5% of the province) as farmland [13].

The "Nan Sandbox" model

A public–private land-management scheme for the province's forest reserve, co-chaired by the Ministry of Natural Resources and the private sector, operating in all 15 districts (924 villages, 2019 figure) [10].



72% remains watershed forest reserve

18% farmland restored to large trees, with crops allowed under them

10% left to economic crops, still legally forest reserve

139,522

farmers cultivating inside reserved or conservation forest land, as counted in the provincial plan [28]

84,429 rai

allocated to communities under the national land policy (CTC), of 124,034 rai approved by the RFD and 273,663 rai requested [29]

301

registered community forests in 14 districts [28]

Sources: [1] NSO Nan Report 2025; [5] Nan Commerce Office, 10 Oct 2025; [8] Zeng, Gower & Wood 2018; [9] Khemphet & Jongkaewwattana 2021; [10] Thai Post, 14 Feb 2019; [12] RFD 2025; [13] Naewna/GISTDA, 22 Aug 2026; [20] OAE, 1 Sep 2023; [21] Kruasilp et al. 2023; [28] Provincial plan, FY2027 review; [29] Nan Agriculture & Cooperatives Office 2024.

Water tower, smoke and floods

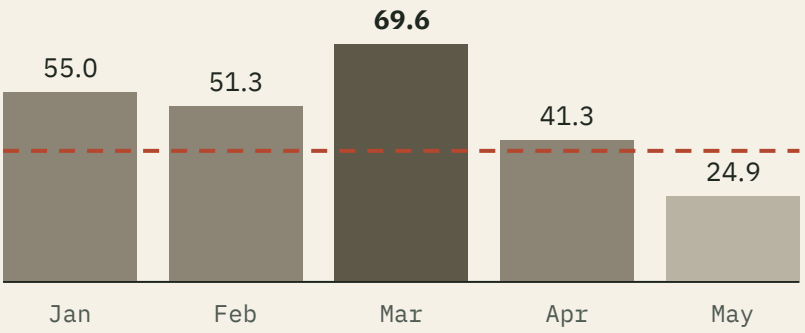
What happens on Nan's slopes is felt downstream: the province heads the main tributary of the Chao Phraya. Locally, dry-season haze and wet-season floods are the recurrent hazards.

≈40%
of the Chao Phraya's runoff comes from the Nan River basin (about 12,000 million m³ a year)

≈57%
of the Chao Phraya's sediment load comes from the same basin

≈34,680 km²
Nan River basin studied, about three times the province's area

PM2.5, monthly average, Jan–May 2025 (µg/m³)



Daily peak: **122.9 µg/m³** on 24 March 2025 — more than three times the national 24-hour standard.

Villages at very high risk, 2024

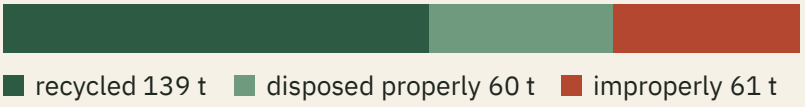
	districts	subdist.	villages
Flood	8	22	**51**
Drought	10	15	**43**
Storm	8	14	**41**

DDPM risk dashboard. Drought is also the second problem cited by farm holders (57.1%, p. 7).

4,847
hotspots a year, avg 2021–25; target −10%

501,738 rai
burned a year, avg 2023–25; target −20%

Solid waste, 2023 — 260 tonnes a day



−29,059 rai

forest change 2015–2025, concentrated in Chaloem Phra Kiat and Thung Chang, the upper Nan watershed

AGRICULTURE & COOPERATIVES PLAN

“Creative, high-value Nan agriculture; farmers happy and sustainable”

Aim: lift farmers out of the middle-income trap. Indicators include agricultural GPP +0.5% and farm-household net income +5% a year.

Threats named in the plan

— Climate change: haze, drought, erratic rain, floods

— Forest and land laws not yet fully implemented

— Steep terrain raising farm costs per rai

Sources: Namsai et al. 2024, Water 16(1):148 [16]; PM2.5 — PM2.5 monitoring system (GISTDA and partners) in Nan Pocket Book 2025 [2]; standard — National Environment Board, ONEP 2022 [19]; risk villages — DDPM dashboard, in Pocket Book [2]; hotspots and burned area — baselines in the provincial plan, FY2027 review [28]; floods — TNN/TrueID 21 & 23 Aug 2026 [14][15], Naewna 22 Aug 2026 [13]; waste — Pollution Control Dept [2]; forest change — RFD data via Naewna/GISTDA [13].

How the province reads its farm sector

The provincial plan's SWOT analysis and the agriculture office's own diagnosis converge on water, land rights, chemical use, ageing farmers and weak processing.

STRENGTHS

— Climate and land suited to high-value products: mango, coffee, longan, orange, tea, herbs;

— Many farmer groups: processing groups, community enterprises, Young Smart Farmers

— Collection centres and processing plants with GMP standards

WEAKNESSES

— Inefficient water management; insecure land documents limit access to production inputs

— Inappropriate chemical use; poor planning along the production chain

— Processing without standards; market information not used, so supply overshoots demand

OPPORTUNITIES

— Growing health consciousness and online, digital sales channels

— The Laos–China railway opens new markets towards China and other countries

— State policies on land allocation (CTC), land titles and royal projects

THREATS

— Climate change bringing plant diseases, animal diseases, pests and disasters

— Epidemics disrupting logistics and markets

— Land and forest laws: farmers without land documents cannot obtain organic or GAP certification

Provincial Development Plan 2023–2027

ISSUE 1 · PEOPLE

Good health, opportunity and a peaceful society

Support people of all ages for good health and a peaceful society.

ISSUE 2 · ECONOMY

A creative, high-value economy

Build infrastructure for farming, water, trade, and tourism

ISSUE 3 · ENVIRONMENT

Environmentally friendly growth

Restore natural resources through public participation.

Selected targets and their baselines

Indicator	Target	Baseline
Life expectancy	**78 years**	77.4 years (2023)
Human achievement index	**≥ 70%**	64% (index 0.6496, 2022)
Population below the poverty line	**down 10%**	11.2% (2020)
Gross provincial product, current prices	**+1.5% a year**	38,938 M THB (2023)
Value of GAP and organic produce	**+4% a year each**	+3% (2021)
Sales of Nan identity products	**+4%**	2,190 M THB (2024)
Forest area, all types	**+0.16%**	+7,536 rai in 2024
Fire hotspots	**–10% vs 5-yr average**	4,847 a year (2021–25)
Burned area	**–20% vs 3-yr average**	501,738 rai a year (2023–25)

Sources: SWOT — Nan Provincial Development Plan 2023–2027, FY2027 review (agriculture) [28]; Nan Provincial Agriculture and Cooperatives Office 2024 [29]; Nan Pocket Book 2025 [2].

Sources

OFFICIAL STATISTICS & PLANS

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[2] NSO Nan (2025). *Nan Pocket Book 2568*.

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Notes on the data

Years are Common Era (Thai B.E. − 543). 1 ha = 6.25 rai; 1 km² = 625 rai.

Crop areas come from farmer registration (DOAE), surveys (OAE), satellite mapping (GISTDA, research) or field estimates (Commerce). Each measures something different, so trends are more reliable than exact levels. Interpolated or estimated values found in preparatory files were not used.

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